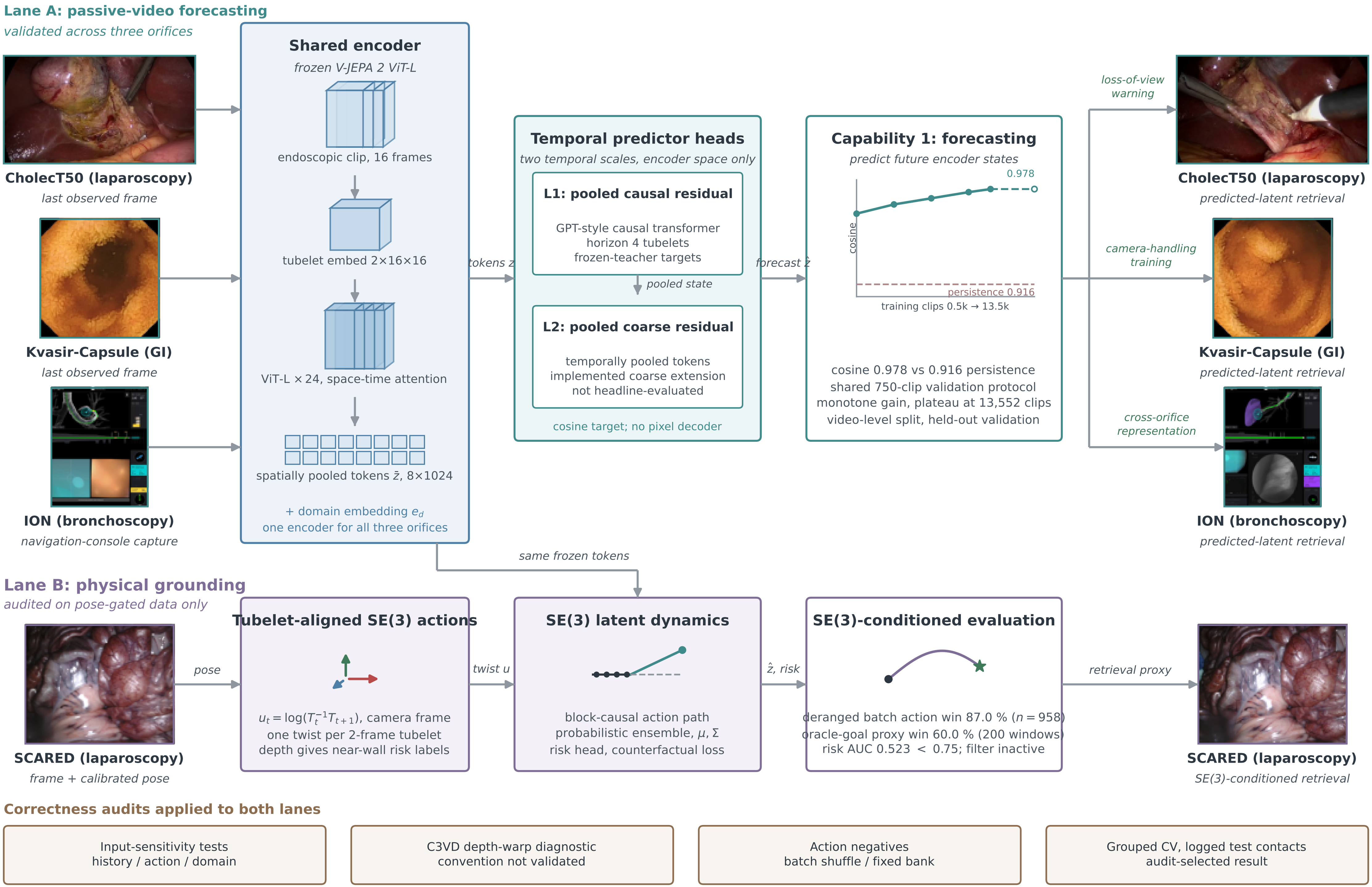

Lane A: passive-video forecasting
validated across three orifices
CholecT50 (laparoscopy)
last observed frame
Kvasir-Capsule (GI)
last observed frame
ION (bronchoscopy)
navigation-console capture
Shared encoder
frozen V-JEPA 2 ViT-L
endoscopic clip, 16 frames
tubelet embed 2×16×16
ViT-L × 24, space-time attention
spatially pooled tokens $\bar{z}$, 8×1024
+ domain embedding e_d
one encoder for all three orifices
tokens $\bar{z}$
Temporal predictor heads
two temporal scales, encoder space only
L1: pooled causal residual
GPT-style causal transformer
horizon 4 tubelets
frozen-teacher targets
pooled state
L2: pooled coarse residual
temporally pooled tokens
implemented coarse extension
not headline-evaluated
cosine target; no pixel decoder
forecast $\hat{z}$
Capability 1: forecasting
predict future encoder states
0.978
cosine
persistence 0.916
training clips 0.5k → 13.5k
cosine 0.978 vs 0.916 persistence
shared 750-clip validation protocol
monotone gain, plateau at 13,552 clips
video-level split, held-out validation
loss-of-view warning
camera-handling training
cross-orifice representation
CholecT50 (laparoscopy)
predicted-latent retrieval
Kvasir-Capsule (GI)
predicted-latent retrieval
ION (bronchoscopy)
predicted-latent retrieval
same frozen tokens
Lane B: physical grounding
audited on pose-gated data only
SCARED (laparoscopy)
frame + calibrated pose
pose
Tubelet-aligned SE(3) actions
$u_t = \log(T_t^{-1} T_{t+1})$, camera frame
one twist per 2-frame tubelet
depth gives near-wall risk labels
twist u
SE(3) latent dynamics
block-causal action path
probabilistic ensemble, μ, Σ
risk head, counterfactual loss
$\hat{z}$, risk
SE(3)-conditioned evaluation
deranged batch action win 87.0 % ($n = 958$)
oracle-goal proxy win 60.0 % (200 windows)
risk AUC 0.523 < 0.75; filter inactive
retrieval proxy
SCARED (laparoscopy)
SE(3)-conditioned retrieval
Correctness audits applied to both lanes
Input-sensitivity tests
history / action / domain
C3VD depth-warp diagnostic
convention not validated
Action negatives
batch shuffle / fixed bank
Grouped CV, logged test contacts
audit-selected result